

Story of How POSTECH Defeats KAIST

Mathematics for Science Quiz

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POSTECH–KAIST Science War
Pohang University of Science and Technology
vs
Korea Advanced Institute of Science and Technology

Preface

This book is devoted to the Mathematics section of Science Quiz in the **POSTECH**–**KAIST** Science War. It collects mathematical concepts, facts to memorize, past problems, and expected problems that are useful for competition preparation.

The main purpose of this text is not merely to record solutions, but to build a practical and reliable mathematical toolkit for Science Quiz. It includes standard theorems, important terminology, memorable facts about mathematicians and mathematical history, and problem-solving strategies that may be useful in an actual match.

I sincerely hope that this book contributes to **POSTECH**'s victory. Good luck, and let the game begin.

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Notation and Conventions

Throughout this book, we use standard mathematical notation whenever possible. The conventions below are arranged to keep the exposition readable, unified, and consistent with the chapter structure of Part I. The guiding principle is simple: use official, standard notation first, and introduce special notation only when it makes a quiz solution faster or clearer.

General Conventions

Scalars are written in ordinary italic letters, such as a, b, c, λ, t . Vectors are written in bold lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Matrices are written in bold uppercase letters, such as $\mathbf{A}, \mathbf{B}, \mathbf{P}$. Unless otherwise stated, vectors are regarded as column vectors. Named theorems, standard English terminology, and conventional symbols are preferred over ad hoc notation.

Notation	Meaning
$\mathbb{N}$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{C}$	the set of complex numbers
$A \subseteq B$	A is a subset of B
$A \subsetneq B$	A is a proper subset of B
$A \setminus B$	the set difference of A and B
$ A $	the cardinality of a finite set A
$\emptyset$	the empty set
λ	usually a scalar, eigenvalue, or parameter, depending on context
ε	a small positive real number, especially in analysis
$\sim$	may mean asymptotic equivalence, distribution, or an equivalence relation
$O(\cdot)$	Big-O notation
$o(\cdot)$	little-o notation
$\square$	end of a proof or solution

Theorem-like Environments

Part I uses theorem-like labels according to the mathematical role of each item. This is especially useful for Science Quiz preparation: a player should be able to tell at a glance whether an item is a definition, a named result, a formula to memorize, or an open problem. Statement-type environments such as Definition and Theorem put the heading on its own line, while short

guide-type environments such as Idea, Strategy, Tactic, and Remark continue on the same line after the label.

Label	Usage
Definition	a precise meaning of a term or object
Theorem	a major named result, usually worth memorizing by name
Proposition	a useful standard result that is less central than a theorem
Lemma	a supporting result or technical tool
Corollary	a result that follows quickly from a theorem or proposition
Formula	a computational identity or closed form that should be recognized quickly
Fact	a memorization-oriented mathematical or historical fact
Conjecture	a statement believed to be true but not proved in full generality
Hypothesis	a named assumption or guiding claim, often attached to a major open problem
Open Problem	a famous unsolved problem or a problem whose general form remains open
Idea	a short conceptual explanation of why a result is true or useful
Strategy	a fast quiz-solving method or recognition trick
Tactic	a local, immediately usable trick for a particular type of question
Remark	supplementary context, caution, or memorable information

Calculus and Vector Calculus

Calculus notation includes one-variable calculus, multivariable calculus, and vector calculus. The symbol ∇ is read as “nabla” or “del.”

Notation	Meaning
$f'(a)$	the derivative of f at a
$f^{(n)}(a)$	the n th derivative of f at a
$\frac{dy}{dx}$	the derivative of y with respect to x
$\int_a^b f(x) dx$	the definite integral of f from a to b
$C^k(I)$	the set of k times continuously differentiable functions on I
$C^\infty(I)$	the set of smooth functions on I
∇f or $\text{grad } f$	the gradient of a scalar-valued function f
$\nabla \cdot \mathbf{F}$ or $\text{div } \mathbf{F}$	the divergence of a vector field $\mathbf{F}$
$\nabla \times \mathbf{F}$ or $\text{curl } \mathbf{F}$	the curl of a vector field $\mathbf{F}$
Δf	the Laplacian of f , usually $\nabla \cdot \nabla f$
$D\mathbf{F}$	the derivative matrix, or Jacobian matrix, of a map $\mathbf{F}$
$\text{Jac}(\mathbf{F})$	the Jacobian determinant of $\mathbf{F}$, when defined
$\int_C \mathbf{F} \cdot d\mathbf{r}$	a line integral of a vector field along a curve C
$\iint_S \mathbf{F} \cdot \mathbf{n} dS$	a flux integral across a surface S

Linear Algebra

In linear algebra, we consistently use bold notation for vectors and matrices. For example, $\mathbf{v} \in \mathbb{R}^n$ is a vector and $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a matrix. The scalar λ is an eigenvalue, not a vector.

Notation	Meaning
$\mathbf{A}^\top$	the transpose of $\mathbf{A}$
$\mathbf{I}_n$	the $n \times n$ identity matrix
$\mathbf{0}$	the zero vector or zero matrix, depending on context
$\det(\mathbf{A})$	the determinant of $\mathbf{A}$
$\text{rk}(\mathbf{A})$	the rank of $\mathbf{A}$
$\text{tr}(\mathbf{A})$	the trace of $\mathbf{A}$
$\text{adj}(\mathbf{A})$	the adjugate of $\mathbf{A}$
$\text{Col}(\mathbf{A})$	the column space of $\mathbf{A}$
$\text{Null}(\mathbf{A})$	the null space of $\mathbf{A}$
$\text{Col}(\mathbf{A}^\top)$	the row space of $\mathbf{A}$
$\text{Row}(\mathbf{A})$	another notation for the row space, used only when convenient
$\text{Ker}(T)$	the kernel of a linear map T
$\text{Im}(T)$	the image of a linear map T
$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	the span of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$
$\langle \mathbf{u}, \mathbf{v} \rangle$	the inner product of $\mathbf{u}$ and $\mathbf{v}$
$\ \mathbf{v}\ $	the norm of $\mathbf{v}$
$\mathbf{u} \perp \mathbf{v}$	$\mathbf{u}$ and $\mathbf{v}$ are orthogonal
$U \oplus V$	the direct sum of subspaces U and V

The four fundamental subspaces of $\mathbf{A} \in \mathbb{R}^{m \times n}$ are written as

$$\text{Col}(\mathbf{A}), \quad \text{Null}(\mathbf{A}), \quad \text{Col}(\mathbf{A}^\top), \quad \text{Null}(\mathbf{A}^\top).$$

We use the standard orthogonal decompositions

$$\mathbb{R}^m = \text{Col}(\mathbf{A}) \oplus \text{Null}(\mathbf{A}^\top), \quad \mathbb{R}^n = \text{Col}(\mathbf{A}^\top) \oplus \text{Null}(\mathbf{A}).$$

For cofactors, we write A_{ij} for the scalar cofactor of the entry a_{ij} of $\mathbf{A}$. The cofactor matrix is denoted by

$$\mathbf{C} = (A_{ij}),$$

and the adjugate matrix is

$$\text{adj}(\mathbf{A}) = \mathbf{C}^\top.$$

We avoid using $\mathbf{A}^*$ for the adjugate, since $\mathbf{A}^*$ often denotes the conjugate transpose in other contexts.

Differential Equations

Differential equations are written using standard derivative notation. The independent variable is usually t or x , depending on the context.

Notation	Meaning
y'	the first derivative of y with respect to the relevant variable

Notation	Meaning
y''	the second derivative of y with respect to the relevant variable
$\dot{x}$	the derivative of x with respect to time t
$\ddot{x}$	the second derivative of x with respect to time t
y_h	the homogeneous part of a solution
y_p	a particular solution
$c_1, c_2, \dots$	arbitrary constants in a general solution
$\mathcal{L}\{f(t)\}(s)$	the Laplace transform of $f(t)$

Combinatorics, Discrete Mathematics, Probability and Statistics

Combinatorics and discrete mathematics provide the notation for counting, recurrence relations, and graphs. Probability and statistics use this notation frequently, especially in finite probability spaces.

Notation	Meaning
$n!$	factorial
$\binom{n}{k}$	the number of k -element subsets of an n -element set
$P(n, k)$	the number of k -permutations of an n -element set
a_n	the n th term of a sequence
$G = (V, E)$	a graph with vertex set V and edge set E
$\deg(v)$	the degree of a vertex v in a graph
K_n	the complete graph on n vertices
C_n	the cycle graph on n vertices
P_n	the path graph on n vertices
$\mathbb{P}(A)$	the probability of an event A
$\mathbb{P}(A \mid B)$	the conditional probability of A given B
$\mathbb{E}[X]$	the expectation of a random variable X
$\text{Var}(X)$	the variance of X
$\text{Cov}(X, Y)$	the covariance of X and Y
$X \sim \text{Bin}(n, p)$	X follows a binomial distribution with parameters n and p
$X \sim \text{Poi}(\lambda)$	X follows a Poisson distribution with parameter λ
$X \sim \text{N}(\mu, \sigma^2)$	X follows a normal distribution with mean μ and variance σ^2

Random variables are usually denoted by uppercase letters such as X, Y, Z . Events are denoted by uppercase letters such as A, B, E . In graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Geometry

We use standard geometric notation unless otherwise stated.

Notation	Meaning
$\triangle ABC$	the triangle with vertices A, B, C
$\angle ABC$	the angle with vertex B
$\overline{AB}$	the line segment joining A and B

Notation	Meaning
AB	the length of the segment $\overline{AB}$, when no confusion is possible
$[ABC]$	the area of triangle ABC , when used in geometry problems
$\parallel$	parallel
$\perp$	perpendicular

Number Theory and Algebra

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$. In algebra, the notation is kept standard and lightweight, since Science Quiz rarely requires highly specialized algebraic formalism.

Notation	Meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$\gcd(a, b)$	the greatest common divisor of a and b
$\text{lcm}(a, b)$	the least common multiple of a and b
$a \equiv b \pmod{n}$	a is congruent to b modulo n
$\varphi(n)$	Euler's phi function, or Euler's totient function
$\text{ord}_n(a)$	the order of a modulo n
p	usually a prime number, when used in number-theoretic contexts
G	usually a group
e	the identity element of a group, when the context is clear
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
G/H	the quotient group of G by H
R	usually a ring
F or K	usually a field
$F[x]$	the polynomial ring over a field F
$\deg f$	the degree of a polynomial f
$\text{Aut}(G)$	the automorphism group of G
$\text{Gal}(L/K)$	the Galois group of a field extension L/K

The Chinese Remainder Theorem may be abbreviated as CRT after it is first introduced. In algebraic contexts, $\deg f$ denotes the degree of a polynomial f ; in graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Miscellaneous Topics

Some symbols have different meanings in different areas. In this book, the intended meaning should always be clear from context.

Notation	Meaning
$\lfloor x \rfloor$	the floor of x
$\lceil x \rceil$	the ceiling of x
$\{x\}$	the fractional part of x , usually $x - \lfloor x \rfloor$

Notation	Meaning
RH	the Riemann Hypothesis, after it has been introduced
P vs NP	the P versus NP problem, after it has been introduced

Part I

Mathematical Concepts

Chapter 1

Calculus

This chapter collects quiz-friendly results from one-variable calculus, multivariable calculus, and vector calculus. The emphasis is on named theorems, memorable formulas, and geometric ideas that are easy to ask and useful to recognize quickly.

One-Variable Calculus

Formula 1.1. Standard Trigonometric and Exponential Limits

The following limits are fundamental:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1, & \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \frac{1}{2}, \\ \lim_{x \rightarrow 0} \frac{e^x - 1}{x} &= 1, & \lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} &= 1,\end{aligned}$$

and

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

Strategy. These limits are often faster than L'Hopital's Rule. In a quiz, the expressions $\sin x$, $1 - \cos x$, $e^x - 1$, $\log(1 + x)$, and $(1 + x)^{1/x}$ are strong signals for these standard patterns.

Theorem 1.2. Intermediate Value Theorem

Let f be continuous on $[a, b]$. If N is any number between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that

$$f(c) = N.$$

Strategy. The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values cross the desired level.

Theorem 1.3. Extreme Value Theorem

Let f be continuous on $[a, b]$. Then f attains both a maximum and a minimum on $[a, b]$.

Remark. The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

Theorem 1.4. Mean Value Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Idea. The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

Theorem 1.5. Taylor's Theorem

Suppose that f is $n + 1$ times differentiable near a . Then

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x),$$

where, in Lagrange's form of the remainder,

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$$

for some c between a and x .

Strategy. Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving e^x , $\sin x$, $\cos x$, $\log(1 + x)$, or $\arctan x$ often invite Taylor expansion.

Formula 1.6. Maclaurin Series

The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1),$$

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \quad (-1 < x \leq 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad (|x| \leq 1).$$

Formula 1.7. Euler's Formula

For every real number x ,

$$e^{ix} = \cos x + i \sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

Remark. The identity $e^{i\pi} + 1 = 0$ is called Euler's identity. It connects the constants e , i , π , 1 , and 0 in a single formula.

Formula 1.8. Gaussian Integral

The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Equivalently,

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Remark. The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

Formula 1.9. Wallis Product

The Wallis product is

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots.$$

Definition 1.10. Gamma Function

For $x > 0$, the Gamma function is defined by

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt.$$

Proposition 1.11. Gamma Function and Factorials

The Gamma function satisfies

$$\Gamma(x+1) = x\Gamma(x).$$

In particular, for every positive integer n ,

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Formula 1.12. Stirling's Formula

As $n \rightarrow \infty$,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

Remark. Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

Multivariable Calculus and Vector Calculus**Definition 1.13. Gradient, Divergence, Curl, and Laplacian**

For a scalar-valued function $f(x, y, z)$ and a vector field $\mathbf{F} = (P, Q, R)$, we write

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right),$$

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix},$$

and

$$\Delta f = \nabla \cdot \nabla f = f_{xx} + f_{yy} + f_{zz}.$$

Idea. The gradient points in the direction of steepest increase. Divergence measures source or sink behavior. Curl measures local rotation. The Laplacian measures how a value differs from its surrounding average.

Theorem 1.14. Clairaut's Theorem

If f has continuous second partial derivatives near a point, then the mixed partial derivatives agree at that point:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Definition 1.15. Jacobian

For a map

$$\mathbf{F}(u_1, \dots, u_n) = (x_1(u_1, \dots, u_n), \dots, x_n(u_1, \dots, u_n)),$$

the Jacobian matrix is

$$D\mathbf{F} = \left(\frac{\partial x_i}{\partial u_j} \right)_{i,j=1}^n.$$

Its determinant

$$\text{Jac}(\mathbf{F}) = \det(D\mathbf{F})$$

is called the Jacobian determinant.

Theorem 1.16. Change of Variables Formula

Under suitable regularity and one-to-one assumptions,

$$\int_{\mathbf{F}(D)} f(\mathbf{x}) d\mathbf{x} = \int_D f(\mathbf{F}(\mathbf{u})) |\det D\mathbf{F}(\mathbf{u})| d\mathbf{u}.$$

Remark. The Jacobian determinant is the local area or volume scaling factor of a change of variables.

Theorem 1.17. Lagrange Multipliers

Suppose that f has a local maximum or minimum subject to the constraint

$$g(x_1, \dots, x_n) = 0$$

at a point where $\nabla g \neq \mathbf{0}$. Then there exists a scalar λ such that

$$\nabla f = \lambda \nabla g.$$

Idea. At a constrained optimum, the level surface of f is tangent to the constraint surface. Thus their normal vectors are parallel.

Theorem 1.18. Green's Theorem

Let C be a positively oriented, simple closed curve in the plane, and let D be the region enclosed by C . Then

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Idea. Green's Theorem converts circulation along a boundary curve into curl over the region inside. It is the two-dimensional ancestor of Stokes' Theorem.

Theorem 1.19. Stokes' Theorem

Let S be an oriented smooth surface with positively oriented boundary curve ∂S . Then

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS.$$

Idea. Stokes' Theorem says that circulation around the boundary equals total curl through the surface. It turns a boundary integral into an interior integral.

Theorem 1.20. Divergence Theorem

Let E be a solid region in $\mathbb{R}^3$ with outward-oriented boundary surface ∂E . Then

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \nabla \cdot \mathbf{F} dV.$$

Idea. The Divergence Theorem says that outward flux through the boundary equals total source strength inside the region.

Remark. Green's Theorem, Stokes' Theorem, and the Divergence Theorem share the same philosophy: a boundary integral is equal to an interior integral. This is the geometric heart of vector calculus.

Chapter 2

Linear Algebra

This chapter focuses on named theorems, matrix invariants, and geometric principles that are especially likely to appear in quiz-style questions.

Theorem 2.1. Rank–Nullity Theorem

Let $T : V \rightarrow W$ be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \operatorname{Ker}(T) + \dim \operatorname{Im}(T).$$

For a matrix $\mathbf{A} \in F^{m \times n}$, this becomes

$$n = \operatorname{rk}(\mathbf{A}) + \dim \operatorname{Null}(\mathbf{A}).$$

Strategy. Rank–Nullity connects variables, pivots, and free variables. It is the linear algebra version of accounting: nothing disappears; dimensions are redistributed.

Theorem 2.2. Invertible Matrix Theorem

For a square matrix $\mathbf{A} \in F^{n \times n}$, the following statements are equivalent:

$$\mathbf{A} \text{ is invertible,} \quad \det(\mathbf{A}) \neq 0, \quad \operatorname{rk}(\mathbf{A}) = n,$$

$$\operatorname{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

has a unique solution for every $\mathbf{b} \in F^n$.

Remark. This theorem is a dictionary. It translates between determinants, rank, null spaces, and solvability of linear systems.

Definition 2.3. Determinant

The determinant is a scalar associated to a square matrix $\mathbf{A} \in F^{n \times n}$, denoted by $\det(\mathbf{A})$. Geometrically, for real matrices, $|\det(\mathbf{A})|$ is the volume scaling factor of the linear transformation $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$.

Proposition 2.4. Basic Properties of Determinants

For square matrices $\mathbf{A}, \mathbf{B} \in F^{n \times n}$, we have

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}),$$

$$\det(\mathbf{A}^T) = \det(\mathbf{A}),$$

and, if $\mathbf{A}$ is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Formula 2.5. Determinant of a Triangular Matrix

If $\mathbf{A}$ is upper triangular or lower triangular with diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$, then

$$\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}.$$

Proposition 2.6. Trace, Determinant, and Eigenvalues

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ have eigenvalues $\lambda_1, \dots, \lambda_n$, counted with algebraic multiplicity. Then

$$\operatorname{tr}(\mathbf{A}) = \lambda_1 + \cdots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Definition 2.7. Eigenvalue and Characteristic Polynomial

Let $\mathbf{A} \in F^{n \times n}$. A scalar $\lambda \in F$ is an eigenvalue of $\mathbf{A}$ if there exists a nonzero vector $\mathbf{v} \in F^n$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

The characteristic polynomial of $\mathbf{A}$ is

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}).$$

Proposition 2.8. Eigenvalues and the Characteristic Polynomial

A scalar λ is an eigenvalue of $\mathbf{A}$ if and only if

$$\det(\lambda\mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial $p_{\mathbf{A}}(\lambda)$.

Theorem 2.9. Cayley–Hamilton Theorem

Every square matrix satisfies its own characteristic polynomial. That is, if

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}),$$

then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

Remark. Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

Theorem 2.10. Diagonalization Criterion

A matrix $\mathbf{A} \in F^{n \times n}$ is diagonalizable over F if and only if F^n has a basis consisting of eigenvectors of $\mathbf{A}$.

Corollary 2.11. Distinct Eigenvalues Imply Diagonalizability

If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues in F , then $\mathbf{A}$ is diagonalizable over F .

Strategy. Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

Theorem 2.12. Spectral Theorem for Real Symmetric Matrices

If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, then all eigenvalues of $\mathbf{A}$ are real, and there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\top}.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Remark. In quiz settings, the words “real symmetric matrix” are a strong signal for real eigenvalues and orthogonal diagonalization.

Theorem 2.13. Cauchy–Schwarz Inequality

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Equivalently, for real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Strategy. The expression $\sum a_i b_i$ is a strong signal for Cauchy–Schwarz. The vector form is often the fastest way to recognize the inequality.

Corollary 2.14. Triangle Inequality

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Theorem 2.15. Gram–Schmidt Process

From any linearly independent list

$$\mathbf{v}_1, \dots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \dots, \mathbf{q}_k$$

with the same span:

$$\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

Remark. Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

Formula 2.16. Projection Matrix

If the columns of $\mathbf{Q} \in \mathbb{R}^{n \times k}$ are orthonormal, then the orthogonal projection matrix onto $\text{Col}(\mathbf{Q})$ is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top.$$

More generally, if the columns of $\mathbf{A} \in \mathbb{R}^{n \times k}$ are linearly independent, then the orthogonal projection matrix onto $\text{Col}(\mathbf{A})$ is

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top.$$

Proposition 2.17. Projection Matrix Criterion

A matrix $\mathbf{P}$ is a projection matrix if

$$\mathbf{P}^2 = \mathbf{P}.$$

It is an orthogonal projection matrix if, in addition,

$$\mathbf{P}^\top = \mathbf{P}.$$

Definition 2.18. Singular Values

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^\top \mathbf{A}.$$

Theorem 2.19. Singular Value Decomposition

Every real matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be written as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top,$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices and $\mathbf{\Sigma}$ is a rectangular diagonal matrix whose diagonal entries are the singular values of $\mathbf{A}$.

Remark. Singular Value Decomposition, or SVD, is one of the most powerful decompositions in linear algebra. It applies to every real matrix, not only to square or diagonalizable matrices.

Theorem 2.20. Jordan Canonical Form

Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every $\mathbf{A} \in \mathbb{C}^{n \times n}$, there exists an invertible matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

is in Jordan canonical form.

Remark. Jordan canonical form is usually more important as a named concept than as a computational tool in Science Quiz. It describes what remains when a matrix cannot be fully diagonalized.

Chapter 3

Differential Equations

This chapter collects named principles and famous model equations from ordinary and partial differential equations.

Theorem 3.1. Picard–Lindelof Existence and Uniqueness Theorem

Consider the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0.$$

If f and $\partial f / \partial y$ are continuous near (t_0, y_0) , then there exists a unique solution near $t = t_0$.

Remark. In a quiz setting, the important phrase is “existence and uniqueness.” It tells us when an initial condition determines exactly one solution.

Theorem 3.2. Superposition Principle

If $y_1, \dots, y_k$ are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1 y_1 + \dots + c_k y_k$$

is also a solution.

Remark. The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

Proposition 3.3. General Solution of a Nonhomogeneous Linear Equation

For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is one particular solution of the nonhomogeneous equation.

Formula 3.4. Second-Order Constant-Coefficient Equation

Consider the homogeneous equation

$$ay'' + by' + cy = 0, \quad a \neq 0.$$

Its characteristic equation is

$$ar^2 + br + c = 0.$$

If the roots are distinct real numbers r_1, r_2 , then

$$y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

If the equation has a repeated real root r , then

$$y_h = (c_1 + c_2 t) e^{rt}.$$

If the roots are $\alpha \pm i\beta$ with $\beta \neq 0$, then

$$y_h = e^{\alpha t} (c_1 \cos \beta t + c_2 \sin \beta t).$$

Remark. The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

Definition 3.5. Wronskian

For two differentiable functions y_1 and y_2 , their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Proposition 3.6. Wronskian and Linear Independence

If

$$W(y_1, y_2)(t_0) \neq 0$$

for some t_0 , then y_1 and y_2 are linearly independent.

Formula 3.7. Exponential Growth and Decay

The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If $k > 0$, this models exponential growth. If $k < 0$, this models exponential decay.

Formula 3.8. Logistic Equation

The logistic equation is

$$y' = ky \left(1 - \frac{y}{K}\right),$$

where $k > 0$ is the growth rate and $K > 0$ is the carrying capacity. Its equilibrium solutions are

$$y = 0 \quad \text{and} \quad y = K.$$

Remark. The logistic equation is the standard model for population growth with limited resources. The parameter K represents the limiting population.

Formula 3.9. Simple Harmonic Oscillator

The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A \cos(\omega t - \phi)$$

for suitable constants A and ϕ .

Remark. This is the equation behind ideal mass-spring motion and many small oscillation models in physics.

Formula 3.10. Damped Harmonic Oscillator

The equation

$$mx'' + cx' + kx = 0$$

models a damped harmonic oscillator. Its characteristic equation is

$$mr^2 + cr + k = 0.$$

The motion is underdamped, critically damped, or overdamped according as

$$c^2 - 4mk < 0, \quad c^2 - 4mk = 0, \quad c^2 - 4mk > 0.$$

Definition 3.11. Laplace Transform

The Laplace transform of a function $f(t)$ is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t) dt.$$

Proposition 3.12. Basic Laplace Transforms

The most common Laplace transforms are

$$\begin{aligned} \mathcal{L}\{1\}(s) &= \frac{1}{s}, & \mathcal{L}\{e^{at}\}(s) &= \frac{1}{s-a}, \\ \mathcal{L}\{\sin at\}(s) &= \frac{a}{s^2 + a^2}, & \mathcal{L}\{\cos at\}(s) &= \frac{s}{s^2 + a^2}. \end{aligned}$$

Remark. The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

Definition 3.13. Heat Equation

The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \quad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

Definition 3.14. Wave Equation

The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed c .

Theorem 3.15. d'Alembert's Formula

The solution of the wave equation

$$u_{tt} = c^2 u_{xx}$$

on the real line with initial conditions

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x),$$

is

$$u(x, t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds.$$

Remark. d'Alembert's formula shows that waves move along the characteristic lines $x - ct = \text{constant}$ and $x + ct = \text{constant}$.

Definition 3.16. Laplace Equation

Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

Definition 3.17. Poisson Equation

Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

Remark. The heat equation, wave equation, and Laplace equation are three of the most famous PDEs. A quiz problem may ask for their names, forms, or physical interpretations.

Chapter 4

Combinatorics and Discrete Mathematics

This chapter collects high-yield results from combinatorics and discrete mathematics. The emphasis is on named principles, memorable formulas, and results with clear quiz-style answers.

Formula 4.1. Binomial Theorem; Newton's Binomial Formula

For every nonnegative integer n ,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

In particular,

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k.$$

Remark. The coefficients $\binom{n}{k}$ are the entries of Pascal's triangle. This is one of the most recognizable bridges between algebra and counting.

Theorem 4.2. Pigeonhole Principle; Dirichlet's Box Principle

If $n + 1$ objects are placed into n boxes, then at least one box contains at least two objects. More generally, if N objects are placed into k boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Strategy. The Pigeonhole Principle is often hidden in problems asking us to prove that two objects must share a property. The hard part is usually choosing the right “boxes.”

Theorem 4.3. Inclusion–Exclusion Principle; Sieve Formula

For finite sets $A_1, \dots, A_n$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|.$$

Remark. Inclusion–Exclusion is the standard correction mechanism for overcounting. It is also historically connected to sieve methods.

Formula 4.4. Derangement Formula

Let D_n be the number of derangements of n objects, that is, permutations with no fixed points. Then

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Moreover,

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

Remark. The approximation $D_n \approx n!/e$ is very memorable. In probability language, a random permutation has no fixed point with probability approximately $1/e$.

Formula 4.5. Catalan Numbers

The n th Catalan number is

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

It begins as

$$1, 1, 2, 5, 14, 42, 132, \dots$$

Remark. Catalan numbers count many different objects, including correct parenthesizations, Dyck paths, triangulations of a polygon, and rooted full binary trees. The number 42 is a classic giveaway.

Theorem 4.6. Cayley's Formula

The number of labeled trees on n vertices is

$$n^{n-2}.$$

Remark. Cayley's formula is a perfect Science Quiz result: the statement is short, the answer is surprising, and the name is famous.

Theorem 4.7. Burnside's Lemma; Cauchy–Frobenius Lemma

Suppose a finite group G acts on a finite set X . The number of orbits is

$$\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|,$$

where

$$\text{Fix}(g) = \{x \in X : g \cdot x = x\}.$$

Idea. Burnside's Lemma says that the number of essentially different objects is the average number of objects fixed by a symmetry.

Lemma 4.8. Handshaking Lemma

For a finite undirected graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

In particular, the number of vertices of odd degree is even.

Remark. The name comes from the idea that every handshake contributes 2 to the total count of hands shaken.

Theorem 4.9. Euler's Formula for Planar Graphs

If a connected planar graph has V vertices, E edges, and F faces, then

$$V - E + F = 2.$$

Remark. This is the graph-theoretic version of Euler's polyhedron formula. It is one of the most famous formulas involving planar graphs.

Theorem 4.10. Eulerian Trail Criterion

A connected finite graph has an Eulerian circuit if and only if every vertex has even degree. It has an Eulerian trail but not an Eulerian circuit if and only if exactly two vertices have odd degree.

Remark. This theorem is historically connected to Euler's solution of the Seven Bridges of Königsberg problem.

Theorem 4.11. Hall's Marriage Theorem

Let $G = (X \cup Y, E)$ be a bipartite graph. There is a matching that covers every vertex of X if and only if, for every subset $S \subseteq X$,

$$|N(S)| \geq |S|,$$

where $N(S)$ is the set of neighbors of vertices in S .

Strategy. Hall's Marriage Theorem is the standard named theorem for proving that a perfect assignment or matching exists.

Theorem 4.12. Ramsey's Theorem

For any positive integers r and s , there exists an integer $R(r, s)$ such that every red-blue coloring of the edges of a sufficiently large complete graph contains either a red K_r or a blue K_s .

Formula 4.13. The Party Problem

The classical Ramsey number

$$R(3, 3) = 6$$

says that among any six people, there are either three mutual acquaintances or three mutual strangers.

Remark. Ramsey theory is the mathematics of unavoidable structure. Its slogan is that complete disorder is impossible in a large enough system.

Theorem 4.14. Erdős–Szekeres Theorem

Any sequence of

$$(r - 1)(s - 1) + 1$$

distinct real numbers contains either an increasing subsequence of length r or a decreasing subsequence of length s .

Remark. This is a beautiful pigeonhole-style result. It is often remembered as the monotone subsequence theorem.

Chapter 5

Probability and Statistics

This chapter focuses on probability results that are easy to ask in a quiz: named theorems, famous distributions, memorable approximations, and inequalities with clear meanings.

Theorem 5.1. Law of Total Probability

If $B_1, \dots, B_n$ form a partition of the sample space and $\mathbb{P}(B_i) > 0$ for all i , then

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

Theorem 5.2. Bayes' Theorem

If $B_1, \dots, B_n$ form a partition of the sample space and $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}.$$

In the two-event form,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B) \mathbb{P}(B)}{\mathbb{P}(A)}.$$

Remark. Bayes' Theorem is the standard formula for reversing conditional probability. It updates a prior probability after observing evidence.

Proposition 5.3. Linearity of Expectation

For random variables $X_1, \dots, X_n$,

$$\mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n].$$

This holds even when the random variables are not independent.

Strategy. Linearity of expectation is often the fastest way to count an expected number of objects. Introduce indicator random variables and add their expectations.

Formula 5.4. Variance and Covariance Identities

For a random variable X ,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

For random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If X and Y are independent, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Formula 5.5. Binomial Distribution

If $X \sim \text{Bin}(n, p)$, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Its mean and variance are

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

Formula 5.6. Poisson Distribution

If $X \sim \text{Poi}(\lambda)$, then

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Its mean and variance are

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

Theorem 5.7. Poisson Limit Theorem; Law of Small Numbers

If $n \rightarrow \infty$, $p \rightarrow 0$, and $np \rightarrow \lambda$, then

$$\binom{n}{k} p^k (1-p)^{n-k} \rightarrow e^{-\lambda} \frac{\lambda^k}{k!}$$

for each fixed $k \geq 0$.

Remark. This theorem explains why the Poisson distribution models rare events: many trials, small success probability, and finite expected count.

Formula 5.8. Normal Distribution; Gaussian Distribution

A random variable $X \sim N(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The standard normal distribution is $N(0, 1)$.

Formula 5.9. Empirical Rule; 68–95–99.7 Rule

For a normal distribution, approximately

68%

of the mass lies within one standard deviation of the mean,

95%

lies within two standard deviations, and

99.7%

lies within three standard deviations.

Theorem 5.10. Weak Law of Large Numbers

Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ . Then, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) \rightarrow 0$$

as $n \rightarrow \infty$.

Idea. The Law of Large Numbers says that the sample average stabilizes near the true mean when the number of trials becomes large.

Theorem 5.11. Central Limit Theorem

Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ and variance $\sigma^2 > 0$. Then

$$\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to the standard normal distribution $N(0, 1)$.

Remark. The Central Limit Theorem explains why the normal distribution appears so often in nature and statistics.

Theorem 5.12. Markov's Inequality

If X is a nonnegative random variable and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 5.13. Chebyshev's Inequality

If X has mean μ and variance σ^2 , then for every $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

Remark. Markov's inequality uses only the mean, while Chebyshev's inequality uses the variance. Both are probability bounds that work without assuming a normal distribution.

Theorem 5.14. Jensen's Inequality

If φ is convex, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

If φ is concave, the inequality is reversed.

Idea. Jensen's inequality says that, for a convex function, applying the function after averaging gives a value no larger than averaging after applying the function.

Formula 5.15. Birthday Problem Approximation

In a group of n people, assuming 365 equally likely birthdays, the probability that at least two people share a birthday is approximately

$$1 - \exp\left(-\frac{n(n-1)}{730}\right).$$

For $n = 23$, this probability is already greater than $1/2$.

Remark. The birthday problem is famous because the answer is surprisingly small: only 23 people are needed for a better-than-even chance of a shared birthday.

Formula 5.16. Gambler's Ruin

Suppose a gambler starts with i dollars and stops when reaching either 0 dollars or N dollars. If the probability of winning each round is p and $q = 1 - p$, then the probability of reaching N before 0 is

$$\frac{1 - (q/p)^i}{1 - (q/p)^N} \quad (p \neq q).$$

In the fair case $p = q = 1/2$, the probability is

$$\frac{i}{N}.$$

Remark. Gambler's ruin is a classic random walk result. It is a good example where the fair case has a very simple answer.

Chapter 6

Geometry

This chapter collects quiz-friendly results from Euclidean synthetic geometry and analytic geometry. The emphasis is on named theorems, elegant geometric configurations, and formulas with clear memorable answers.

Synthetic Geometry

Definition 6.1. Classical Triangle Centers

For a triangle $\triangle ABC$, the most important classical centers are:

$$O = \text{circumcenter}, \quad I = \text{incenter}, \quad H = \text{orthocenter}, \quad G = \text{centroid}.$$

Here O is the center of the circumcircle, I is the center of the incircle, H is the intersection of the altitudes, and G is the intersection of the medians.

Remark. Triangle centers are extremely common in olympiad-style geometry. The symbols O, I, H, G are standard and should be recognized quickly.

Formula 6.2. Extended Law of Sines

For a triangle $\triangle ABC$ with side lengths

$$a = BC, \quad b = CA, \quad c = AB,$$

and circumradius R , we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Formula 6.3. Heron's Formula

Let $\triangle ABC$ have side lengths a, b, c and semiperimeter

$$s = \frac{a + b + c}{2}.$$

Then its area K is

$$K = \sqrt{s(s-a)(s-b)(s-c)}.$$

Also,

$$K = rs \quad \text{and} \quad K = \frac{abc}{4R},$$

where r is the inradius and R is the circumradius.

Strategy. The formulas

$$K = rs, \quad K = \frac{abc}{4R}$$

are often more useful than Heron's formula when incircles or circumcircles appear.

Theorem 6.4. Angle Bisector Theorem

In a triangle $\triangle ABC$, suppose the internal angle bisector of $\angle A$ meets BC at D . Then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

Theorem 6.5. Ceva's Theorem

Let D, E, F lie on the lines BC, CA, AB , respectively. Then the cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$

using directed lengths.

Remark. Ceva's Theorem is the standard theorem for proving concurrence of three lines in a triangle.

Theorem 6.6. Menelaus' Theorem

Let D, E, F lie on the lines BC, CA, AB , respectively. Then the points D, E, F are collinear if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1,$$

using directed lengths.

Remark. Ceva is for concurrence, while Menelaus is for collinearity. This pair is one of the most important toolkits in olympiad geometry.

Theorem 6.7. Euler Line Theorem

In any non-equilateral triangle, the circumcenter O , centroid G , and orthocenter H are collinear. Moreover,

$$OG : GH = 1 : 2.$$

The line passing through these points is called the Euler line.

Theorem 6.8. Nine-Point Circle Theorem; Euler Circle

In any triangle, the following nine points lie on one circle: the three side midpoints, the three feet of the altitudes, and the three midpoints between the orthocenter and the vertices. This circle is called the nine-point circle.

Remark. The center of the nine-point circle is the midpoint of the segment joining the circumcenter O and the orthocenter H .

Theorem 6.9. Euler's Formula for the Incenter and Circumcenter

For a triangle with circumradius R , inradius r , circumcenter O , and incenter I , we have

$$OI^2 = R^2 - 2Rr.$$

In particular,

$$R \geq 2r.$$

Remark. The inequality $R \geq 2r$ is called Euler's inequality for triangles. Equality holds exactly for an equilateral triangle.

Theorem 6.10. Power of a Point Theorem

Let P be a point and let a line through P meet a circle at A and B . The product

$$PA \cdot PB$$

is independent of the choice of the line through P . It is called the power of P with respect to the circle.

Formula 6.11. Tangent–Secant Theorem

If PT is tangent to a circle at T and a secant through P meets the circle at A and B , then

$$PT^2 = PA \cdot PB.$$

Theorem 6.12. Radical Axis Theorem

For two nonconcentric circles, the set of points having equal powers with respect to the two circles is a line. This line is called the radical axis of the two circles.

Remark. Radical axes are powerful because they turn circle configurations into collinearity statements.

Theorem 6.13. Ptolemy's Theorem

A quadrilateral $ABCD$ is cyclic if and only if

$$AC \cdot BD = AB \cdot CD + BC \cdot DA$$

for the appropriate cyclic order of the vertices.

Remark. Ptolemy's Theorem is one of the most recognizable formulas for cyclic quadrilaterals.

Theorem 6.14. Simson Line Theorem

Let P be a point on the circumcircle of $\triangle ABC$. The feet of the perpendiculars from P to the three lines AB, BC, CA are collinear. This line is called the Simson line of P with respect to $\triangle ABC$.

Theorem 6.15. Miquel's Theorem

In a complete quadrilateral, the circumcircles of the four triangles formed by choosing three of the four lines pass through a common point. This point is called the Miquel point.

Remark. Miquel's Theorem is a standard source of hidden cyclic quadrilaterals in olympiad geometry.

Theorem 6.16. Butterfly Theorem

Let M be the midpoint of a chord PQ of a circle. Through M , draw two other chords AB and CD , and let AD and BC meet PQ at X and Y , respectively. Then M is also the midpoint of XY .

Theorem 6.17. Morley's Trisector Theorem

In any triangle, the adjacent trisectors of the three angles intersect in three points that form an equilateral triangle.

Remark. Morley's theorem is famous because the conclusion is surprisingly simple: angle trisectors always create an equilateral triangle.

Theorem 6.18. Napoleon's Theorem

Construct equilateral triangles externally on the three sides of any triangle. Then the centers of these three equilateral triangles form an equilateral triangle.

Theorem 6.19. Classification of Platonic Solids

There are exactly five Platonic solids:

tetrahedron, cube, octahedron, dodecahedron, icosahedron.

Remark. The classification of Platonic solids is a classic result in Euclidean geometry and a very natural quiz fact.

Analytic Geometry

Formula 6.20. Shoelace Formula; Gauss's Area Formula

For a polygon with vertices

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

in cyclic order, its area is

$$\frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right|,$$

where

$$(x_{n+1}, y_{n+1}) = (x_1, y_1).$$

Remark. The Shoelace Formula is one of the cleanest examples of analytic geometry: a geometric area becomes a determinant-like coordinate calculation.

Definition 6.21. Conic Section

A conic section is a curve obtained by intersecting a plane with a double cone. The nondegenerate conics are:

ellipse, parabola, hyperbola.

Equivalently, a conic can be described using a focus, a directrix, and an eccentricity e .

Formula 6.22. Focus–Directrix Definition of Conics

A conic is the set of points P satisfying

$$\frac{\text{dist}(P, \text{focus})}{\text{dist}(P, \text{directrix})} = e.$$

It is an ellipse if

$$0 < e < 1,$$

a parabola if

$$e = 1,$$

and a hyperbola if

$$e > 1.$$

Formula 6.23. Standard Forms of Conics

The standard form of an ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

The standard form of a hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The standard form of a parabola opening to the right is

$$y^2 = 4px.$$

Formula 6.24. Eccentricity of an Ellipse and a Hyperbola

For an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a \geq b > 0),$$

the eccentricity is

$$e = \frac{c}{a}, \quad c^2 = a^2 - b^2.$$

For a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

the eccentricity is

$$e = \frac{c}{a}, \quad c^2 = a^2 + b^2.$$

Theorem 6.25. Reflection Property of Conics

An ellipse reflects a ray from one focus to the other focus. A parabola reflects rays parallel to its axis through its focus. A hyperbola has the analogous reflection property involving its two foci and branches.

Remark. The reflection property gives a geometric reason why conics appear in optics, satellite dishes, whispering galleries, and orbital mechanics.

Theorem 6.26. Quadratic Classification of Conics

Consider a real quadratic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

The discriminant

$$B^2 - 4AC$$

determines the type of the nondegenerate conic:

$$B^2 - 4AC < 0 \quad \Rightarrow \quad \text{ellipse-type},$$

$$B^2 - 4AC = 0 \quad \Rightarrow \quad \text{parabola-type},$$

and

$$B^2 - 4AC > 0 \quad \Rightarrow \quad \text{hyperbola-type}.$$

Remark. This is the analytic-geometry counterpart of classifying conic sections by their shapes.

Formula 6.27. Polar Equation of a Conic

With a focus at the pole, a conic has polar equation

$$r = \frac{\ell}{1 + e \cos \theta}$$

after choosing a suitable axis. Here e is the eccentricity and ℓ is the semi-latus rectum.

Remark. The same equation describes ellipses, parabolas, and hyperbolas depending on whether $e < 1$, $e = 1$, or $e > 1$.

Definition 6.28. Apollonius Circle

Given two fixed points A and B and a positive constant $k \neq 1$, the locus of points P satisfying

$$\frac{PA}{PB} = k$$

is a circle. This circle is called an Apollonius circle.

Remark. Apollonius circles turn a ratio of distances into a concrete circle, which is a common hidden structure in geometry problems.

Theorem 6.29. Descartes' Circle Theorem; Soddy Circle Theorem

Suppose four circles are mutually tangent, and let their curvatures be

$$k_1, k_2, k_3, k_4,$$

where curvature means the reciprocal of the radius, with sign convention for an enclosing circle. Then

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

Remark. Descartes' Circle Theorem is a striking formula: tangency of four circles becomes one quadratic equation in their curvatures.

Theorem 6.30. Pick's Theorem

Let P be a simple polygon whose vertices lie on lattice points in the plane. If I is the number of lattice points strictly inside P and B is the number of lattice points on the boundary of P , then

$$\text{Area}(P) = I + \frac{B}{2} - 1.$$

Remark. Pick's Theorem is a beautiful bridge between geometry and arithmetic: area is determined by counting lattice points.

Chapter 7

Number Theory

This chapter collects famous number-theoretic results with strong quiz value: primes, congruences, reciprocity, sums of squares, and the distribution of primes.

Theorem 7.1. Fundamental Theorem of Arithmetic

Every integer $n > 1$ can be written as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. Equivalently,

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where $p_1, \dots, p_k$ are distinct primes and $\alpha_1, \dots, \alpha_k$ are positive integers, and this expression is unique up to reordering.

Remark. This theorem is often called unique prime factorization. It is the reason prime numbers are treated as the basic building blocks of integers.

Theorem 7.2. Euclid's Theorem; Infinitude of Primes

There are infinitely many prime numbers.

Idea. If $p_1, \dots, p_n$ are finitely many primes, consider

$$p_1 p_2 \cdots p_n + 1.$$

This number is not divisible by any prime on the list.

Theorem 7.3. Bezout's Identity

Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

In particular, if $\gcd(a, b) = 1$, then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = 1.$$

Remark. Bezout's identity is the algebraic heart of the Euclidean algorithm and modular inverses.

Theorem 7.4. Chinese Remainder Theorem; CRT

Let $m_1, \dots, m_k$ be pairwise relatively prime positive integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \quad x \equiv a_2 \pmod{m_2}, \quad \dots, \quad x \equiv a_k \pmod{m_k}$$

has a solution. Moreover, the solution is unique modulo

$$M = m_1 m_2 \cdots m_k.$$

Strategy. CRT is the standard theorem for combining several congruence conditions into one congruence.

Formula 7.5. Euler's Phi Function Formula

If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

is the prime factorization of n , then

$$\varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

In particular, for a prime power,

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1).$$

Theorem 7.6. Fermat's Little Theorem

Let p be a prime number. If $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a ,

$$a^p \equiv a \pmod{p}.$$

Theorem 7.7. Euler's Theorem

If $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Remark. Fermat's Little Theorem is the special case of Euler's Theorem when $n = p$ is prime, since $\varphi(p) = p - 1$.

Theorem 7.8. Wilson's Theorem

A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Remark. Wilson's theorem is famous because it gives a clean factorial characterization of primes, even though it is not efficient as a primality test.

Definition 7.9. Legendre Symbol

Let p be an odd prime. The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1, & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0, & \text{if } p \mid a. \end{cases}$$

Theorem 7.10. Euler's Criterion

Let p be an odd prime. For any integer a ,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Equivalently, if $p \nmid a$, then

$$a^{(p-1)/2} \equiv \begin{cases} 1 \pmod{p}, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 \pmod{p}, & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

Theorem 7.11. Law of Quadratic Reciprocity; Gauss's Theorem of Quadratic Reciprocity

Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless both p and q are congruent to 3 modulo 4.

Remark. Quadratic reciprocity is one of the crown jewels of elementary number theory. Gauss called it the golden theorem.

Theorem 7.12. Fermat's Two-Square Theorem

An odd prime p can be written as a sum of two squares,

$$p = a^2 + b^2$$

for some integers $a, b \in \mathbb{Z}$, if and only if

$$p \equiv 1 \pmod{4}.$$

Also,

$$2 = 1^2 + 1^2.$$

Theorem 7.13. Lagrange's Four-Square Theorem

Every nonnegative integer can be written as a sum of four integer squares. That is, for every $n \in \mathbb{N}_0$, there exist integers $a, b, c, d \in \mathbb{Z}$ such that

$$n = a^2 + b^2 + c^2 + d^2.$$

Remark. The two-square theorem has a congruence condition. The four-square theorem has no exception at all.

Theorem 7.14. Dirichlet's Theorem on Arithmetic Progressions

If a and d are relatively prime positive integers, then the arithmetic progression

$$a, a + d, a + 2d, a + 3d, \dots$$

contains infinitely many primes.

Remark. Dirichlet's theorem is a major result connecting primes and arithmetic progressions. The condition $\gcd(a, d) = 1$ is necessary.

Theorem 7.15. Bertrand's Postulate; Chebyshev's Theorem

For every integer $n > 1$, there exists a prime p such that

$$n < p < 2n.$$

Theorem 7.16. Prime Number Theorem

Let $\pi(x)$ be the number of primes less than or equal to x . Then

$$\pi(x) \sim \frac{x}{\log x} \quad \text{as } x \rightarrow \infty.$$

Idea. The Prime Number Theorem says that the “probability” that a large integer near x is prime is roughly $1/\log x$. This is a mental model, not a literal probability statement.

Definition 7.17. Riemann Zeta Function

For $\operatorname{Re}(s) > 1$, the Riemann zeta function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It also has the Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Remark. The Euler product is the bridge between the zeta function and prime numbers. The Riemann Hypothesis asserts that the nontrivial zeros of $\zeta(s)$ all have real part $1/2$.

Theorem 7.18. Euclid–Euler Theorem on Perfect Numbers

An even positive integer N is perfect if and only if

$$N = 2^{p-1}(2^p - 1),$$

where $2^p - 1$ is a Mersenne prime.

Remark. A perfect number is a positive integer equal to the sum of its positive proper divisors. For example,

$$6 = 1 + 2 + 3.$$

Theorem 7.19. Fermat's Last Theorem

For every integer $n \geq 3$, there do not exist positive integers a, b, c such that

$$a^n + b^n = c^n.$$

Remark. Fermat's Last Theorem was proved by Andrew Wiles. It is one of the most famous theorems in the history of mathematics.

Chapter 8

Algebra

This chapter collects algebraic structures and named theorems that are likely to appear as terminology or quick-recognition questions. The emphasis is on groups, rings, fields, and polynomial equations, not on long abstract proofs.

Definition 8.1. Group

A group is a set G together with a binary operation satisfying closure, associativity, the existence of an identity element, and the existence of inverses. If the operation is commutative, then G is called an abelian group.

Definition 8.2. Cyclic Group

A group G is cyclic if there exists an element $g \in G$ such that every element of G is a power of g . In this case, we write

$$G = \langle g \rangle.$$

Theorem 8.3. Lagrange's Theorem

If G is a finite group and $H \leq G$, then

$$|H| \mid |G|.$$

Equivalently, the order of every subgroup divides the order of the group.

Corollary 8.4. Order of an Element

If G is a finite group and $g \in G$, then the order of g divides $|G|$. In particular,

$$g^{|G|} = e.$$

Theorem 8.5. Cauchy's Theorem

If G is a finite group and a prime p divides $|G|$, then G contains an element of order p .

Theorem 8.6. First Isomorphism Theorem for Groups

If $\varphi : G \rightarrow H$ is a group homomorphism, then

$$G/\text{Ker}(\varphi) \cong \text{Im}(\varphi).$$

Theorem 8.7. Cayley's Theorem

Every group is isomorphic to a subgroup of a symmetric group. In particular, every finite group of order n is isomorphic to a subgroup of S_n .

Theorem 8.8. Sylow Theorems

Let G be a finite group and suppose

$$|G| = p^k m, \quad p \nmid m,$$

where p is prime. Then G has a subgroup of order p^k , called a Sylow p -subgroup. Moreover, all Sylow p -subgroups are conjugate, and the number n_p of Sylow p -subgroups satisfies

$$n_p \equiv 1 \pmod{p}, \quad n_p \mid m.$$

Strategy. For Science Quiz, the most common finite-group quick check is divisibility: subgroup orders and element orders must divide the group order. When a prime divides $|G|$, Cauchy's theorem guarantees an element of that prime order.

Definition 8.9. Ring and Field

A ring is a set equipped with addition and multiplication satisfying the usual distributive laws. A field is a commutative ring in which every nonzero element has a multiplicative inverse.

Definition 8.10. Integral Domain

An integral domain is a commutative ring with identity in which there are no zero divisors. In other words, if $ab = 0$, then $a = 0$ or $b = 0$.

Theorem 8.11. Polynomial Division Algorithm

Let F be a field. If $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, then there exist unique polynomials $q(x), r(x) \in F[x]$ such that

$$f(x) = q(x)g(x) + r(x), \quad r(x) = 0 \quad \text{or} \quad \deg r < \deg g.$$

Theorem 8.12. Factor Theorem

For a polynomial $f(x) \in F[x]$ and an element $a \in F$,

$$f(a) = 0 \iff x - a \text{ divides } f(x).$$

Theorem 8.13. Fundamental Theorem of Algebra

Every nonconstant complex polynomial has at least one complex root. Equivalently, every polynomial of degree $n \geq 1$ over $\mathbb{C}$ splits into n linear factors over $\mathbb{C}$, counted with multiplicity.

Definition 8.14. Algebraically Closed Field

A field F is algebraically closed if every nonconstant polynomial in $F[x]$ has a root in F .

Theorem 8.15. Galois Correspondence

For a finite Galois extension L/K , there is an inclusion-reversing correspondence between intermediate fields $K \subseteq E \subseteq L$ and subgroups $H \leq \text{Gal}(L/K)$, given by

$$E \longmapsto \text{Gal}(L/E), \quad H \longmapsto L^H.$$

Theorem 8.16. Abel–Ruffini Theorem

There is no formula for the roots of a general polynomial of degree 5 or higher using only the field operations and radicals.

Remark. The Abel–Ruffini theorem does not say that every quintic equation is unsolvable. It says that no universal radical formula exists for the general quintic.

Chapter 9

Analysis

This chapter collects foundational results from real analysis that often appear behind rigorous calculus, limits, compactness, and convergence. The statements are kept theorem-oriented so that they can be recognized quickly in a quiz setting.

Definition 9.1. Limit Superior and Limit Inferior

For a real sequence (a_n) , define

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k, \quad \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k,$$

whenever these extended real limits are allowed.

Theorem 9.2. Monotone Convergence Theorem for Sequences

Every bounded monotone real sequence converges. More precisely, every increasing sequence bounded above converges to its supremum, and every decreasing sequence bounded below converges to its infimum.

Theorem 9.3. Bolzano–Weierstrass Theorem

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 9.4. Heine–Borel Theorem

A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Theorem 9.5. Uniform Continuity on Compact Sets

If $K \subseteq \mathbb{R}^n$ is compact and $f : K \rightarrow \mathbb{R}^m$ is continuous, then f is uniformly continuous on K .

Theorem 9.6. Weierstrass Extreme Value Theorem

If $K \subseteq \mathbb{R}^n$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then f attains both a maximum and a minimum on K .

Strategy. In analysis questions, compactness usually means that one may extract a convergent subsequence or that a continuous function attains extrema. In $\mathbb{R}^n$, the fast recognition phrase is “closed and bounded.”

Theorem 9.7. Cauchy Criterion

A sequence (a_n) in $\mathbb{R}$ or $\mathbb{C}$ converges if and only if it is Cauchy: for every $\varepsilon > 0$, there exists N such that

$$m, n \geq N \implies |a_m - a_n| < \varepsilon.$$

Theorem 9.8. Weierstrass M-Test

Let (f_n) be a sequence of functions on a set E . If

$$|f_n(x)| \leq M_n \quad \text{for all } x \in E$$

and

$$\sum_{n=1}^{\infty} M_n$$

converges, then the series

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on E .

Theorem 9.9. Uniform Limit Theorem

If (f_n) is a sequence of continuous functions on a set E and $f_n \rightarrow f$ uniformly on E , then f is continuous on E .

Theorem 9.10. Term-by-Term Differentiation

Suppose f_n are continuously differentiable on an interval and f_n converges at one point. If f'_n converges uniformly, then f_n converges uniformly to a differentiable function f , and

$$f' = \lim_{n \rightarrow \infty} f'_n.$$

Theorem 9.11. Arzelà–Ascoli Theorem

A uniformly bounded and equicontinuous family of real-valued continuous functions on a compact metric space has a uniformly convergent subsequence.

Theorem 9.12. Banach Fixed-Point Theorem; Contraction Mapping Theorem

Let (X, d) be a complete metric space, and let $T : X \rightarrow X$ be a contraction. Then T has a unique fixed point $x^* \in X$, and the iteration $x_{n+1} = T(x_n)$ converges to x^* for every initial point $x_0 \in X$.

Theorem 9.13. Baire Category Theorem

A complete metric space cannot be written as a countable union of nowhere dense closed sets. Equivalently, a countable intersection of open dense subsets of a complete metric space is dense.

Theorem 9.14. Lebesgue Dominated Convergence Theorem

If $f_n \rightarrow f$ almost everywhere and there exists an integrable function g such that $|f_n| \leq g$ for all n , then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

Theorem 9.15. Fubini's Theorem

Under suitable integrability assumptions, an iterated integral may be computed in either order:

$$\int_X \left(\int_Y f(x, y) dy \right) dx = \int_Y \left(\int_X f(x, y) dx \right) dy.$$

Chapter 10

Topology

This chapter records topology terms and theorems that are famous enough to appear in mathematical culture and Science Quiz questions. The goal is not to develop topology from scratch, but to make the names and meanings recognizable.

Definition 10.1. Topology

A topology on a set X is a collection $\mathcal{T}$ of subsets of X whose elements are called open sets, satisfying:

- (i) $\emptyset, X \in \mathcal{T}$;
- (ii) arbitrary unions of sets in $\mathcal{T}$ belong to $\mathcal{T}$;
- (iii) finite intersections of sets in $\mathcal{T}$ belong to $\mathcal{T}$.

Definition 10.2. Compactness

A topological space X is compact if every open cover of X has a finite subcover.

Definition 10.3. Connectedness

A topological space X is connected if it cannot be written as the union of two disjoint nonempty open sets. Equivalently, the only subsets of X that are both open and closed are $\emptyset$ and X .

Definition 10.4. Homeomorphism

A homeomorphism is a bijective continuous map whose inverse is also continuous. Two spaces are homeomorphic if they are the same from the viewpoint of topology.

Theorem 10.5. Brouwer Fixed-Point Theorem

Every continuous map from a closed ball in $\mathbb{R}^n$ to itself has at least one fixed point.

Theorem 10.6. Invariance of Domain

If $U \subseteq \mathbb{R}^n$ is open and $f : U \rightarrow \mathbb{R}^n$ is continuous and injective, then $f(U)$ is open and f is a homeomorphism from U onto $f(U)$.

Definition 10.7. Manifold

An n -dimensional topological manifold is a space that locally looks like $\mathbb{R}^n$. More precisely, each point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

Definition 10.8. Fundamental Group

The fundamental group $\pi_1(X, x_0)$ records loops in X based at x_0 , up to continuous deformation. It is one of the basic algebraic invariants of a topological space.

Fact 10.9. Fundamental Groups of Familiar Spaces

The following basic examples are worth recognizing:

$$\pi_1(S^1) \cong \mathbb{Z}, \quad \pi_1(S^n) \cong 0 \quad (n \geq 2), \quad \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2.$$

Formula 10.10. Euler Characteristic of a Surface

For a triangulated closed surface, the Euler characteristic is

$$\chi = V - E + F.$$

For an orientable closed surface of genus g ,

$$\chi = 2 - 2g.$$

Theorem 10.11. Classification of Closed Surfaces

Every connected closed surface is homeomorphic to either an orientable surface of genus $g \geq 0$ or a nonorientable surface obtained as a connected sum of $k \geq 1$ projective planes.

Theorem 10.12. Poincaré Conjecture

Every closed simply connected 3-manifold is homeomorphic to the 3-sphere S^3 .

Remark. The Poincaré conjecture was proved by Grigori Perelman using Ricci flow. In a quiz setting, the key association is usually

$$\text{Poincaré conjecture} \longleftrightarrow \text{Perelman} \longleftrightarrow \text{Ricci flow}.$$

Theorem 10.13. Thurston Geometrization Theorem

Every closed orientable 3-manifold can be decomposed into pieces that admit one of the eight Thurston geometries.

Fact 10.14. The Eight Thurston Geometries

The eight model geometries are

$$S^3, \quad \mathbb{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\text{SL}_2(\mathbb{R})}, \quad \text{Nil}, \quad \text{Sol}.$$

Chapter 11

Miscellaneous Topics

Formula 11.1. Mercator Series for $\ln 2$

The alternating harmonic series satisfies

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2.$$

Remark. This follows from the Mercator series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

by substituting $x = 1$.

Formula 11.2. Basel Sum; Euler's Basel Formula

The Basel sum is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}.$$

Equivalently,

$$\zeta(2) = \frac{\pi^2}{6}.$$

Remark. The problem of evaluating

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is called the Basel problem. Its famous solution was found by Euler.

Formula 11.3. Odd Part of the Basel Sum

The sum of reciprocal squares over odd positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

Idea. This is obtained by subtracting the even part from the Basel sum:

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \left(1 - \frac{1}{4}\right) \frac{\pi^2}{6} = \frac{\pi^2}{8}.$$

Formula 11.4. Alternating Basel Series; Dirichlet Eta Value at 2

The alternating Basel series satisfies

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}.$$

Equivalently,

$$\eta(2) = \frac{\pi^2}{12},$$

where $\eta(s)$ is the Dirichlet eta function.

Idea. The positive terms are the odd reciprocal squares and the negative terms are the even reciprocal squares. Hence

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

Formula 11.5. Gregory–Leibniz Series; Leibniz Formula for π

The Gregory–Leibniz series is

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2n-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

Equivalently,

$$\pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right).$$

Remark. This follows from the Taylor series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

by substituting $x = 1$, since $\arctan 1 = \pi/4$.

Strategy. These values are useful because Science Quiz problems often hide them behind a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right) = \frac{\ln 2}{2}.$$

Part II

Facts to Memorize

Chapter 12

ICM and the Fields Medal

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.

Year	ICM location	Medalists and areas	Notes
1936	Oslo, Norway	Lars Ahlfors — complex analysis, Riemann surfaces; Jesse Douglas — minimal surfaces, Plateau problem	First Fields Medals.
1950	Cambridge, USA	Laurent Schwartz — distribution theory; Atle Selberg — analytic number theory, sieve methods, zeta functions	First awards after World War II.
1954	Amsterdam, Netherlands	Kunihiko Kodaira — complex geometry, algebraic geometry; Jean-Pierre Serre — algebraic topology, sheaf theory	Serre is the youngest Fields Medalist.
1958	Edinburgh, UK	Klaus Roth — number theory, Diophantine approximation; René Thom — cobordism, differential topology	
1962	Stockholm, Sweden	Lars Hörmander — partial differential equations; John Milnor — differential topology, exotic spheres	
1966	Moscow, USSR	Michael Atiyah — K-theory, index theorem; Paul Cohen — set theory, forcing, continuum hypothesis; Alexander Grothendieck — algebraic geometry, homological algebra; Stephen Smale — differential topology, generalized Poincaré conjecture	First year with four medalists.
1970	Nice, France	Alan Baker — transcendental number theory; Heisuke Hironaka — algebraic geometry, resolution of singularities; Serge Novikov — topology, differential geometry; John G. Thompson — finite group theory	
1974	Vancouver, Canada	Enrico Bombieri — number theory, complex analysis, minimal surfaces; David Mumford — algebraic geometry, moduli spaces	
1978	Helsinki, Finland	Pierre Deligne — algebraic geometry, Weil conjectures; Charles Fefferman — analysis, several complex variables; Grigory Margulis — Lie groups, ergodic theory, discrete groups; Daniel Quillen — algebraic K-theory	

Continued on the next page.

Year	ICM location	Medalists and areas	Notes
1982	Warsaw, Poland	Alain Connes — operator algebras, noncommutative geometry; William Thurston — low-dimensional topology, 3-manifolds; Shing-Tung Yau — differential geometry, PDE, Calabi conjecture	ICM Warsaw was held in 1983.
1986	Berkeley, USA	Simon Donaldson — four-manifold topology, gauge theory; Gerd Faltings — arithmetic algebraic geometry, Mordell conjecture; Michael Freedman — four-manifold topology, four-dimensional Poincaré conjecture	
1990	Kyoto, Japan	Vladimir Drinfeld — quantum groups, Langlands program; Vaughan Jones — operator algebras, knot theory; Shigefumi Mori — algebraic geometry, minimal model program; Edward Witten — mathematical physics, topology, quantum field theory	Witten was the first physicist to receive the Fields Medal.
1994	Zurich, Switzerland	Jean Bourgain — analysis, Banach spaces, harmonic analysis; Pierre-Louis Lions — partial differential equations; Jean-Christophe Yoccoz — dynamical systems; Efim Zelmanov — algebra, restricted Burnside problem	
1998	Berlin, Germany	Richard Borcherds — algebra, automorphic forms, moonshine; W. Timothy Gowers — functional analysis, combinatorics; Maxim Kontsevich — algebraic geometry, topology, mathematical physics; Curtis T. McMullen — complex dynamics, hyperbolic geometry	Andrew Wiles received a special IMU silver plaque.
2002	Beijing, China	Laurent Lafforgue — Langlands correspondence over function fields; Vladimir Voevodsky — motivic cohomology, $\mathbb{A}^1$ -homotopy theory	
2006	Madrid, Spain	Andrei Okounkov — probability, representation theory, algebraic geometry; Grigori Perelman — Ricci flow, Poincaré conjecture; Terence Tao — PDE, harmonic analysis, additive combinatorics; Wendelin Werner — probability, SLE, Brownian motion	Perelman declined the medal.
2010	Hyderabad, India	Elon Lindenstrauss — ergodic theory, number theory; Ngo Bao Chau — automorphic forms, Fundamental Lemma; Stanislav Smirnov — probability, statistical physics, conformal invariance; Cédric Villani — kinetic theory, Boltzmann equation, optimal transport	
2014	Seoul, South Korea	Artur Avila — dynamical systems; Manjul Bhargava — number theory, geometry of numbers; Martin Hairer — stochastic partial differential equations; Maryam Mirzakhani — Riemann surfaces, moduli spaces, dynamics	Mirzakhani was the first woman Fields Medalist.
2018	Rio de Janeiro, Brazil	Caucher Birkar — algebraic geometry, Fano varieties, minimal model program; Alessio Figalli — optimal transport, PDE, metric geometry; Peter Scholze — arithmetic algebraic geometry, p -adic geometry; Akshay Venkatesh — number theory, homogeneous dynamics, representation theory	

Continued on the next page.

Year	ICM location	Medalists and areas	Notes
2022	Helsinki, Finland	Hugo Duminil-Copin — probability, statistical physics, phase transitions; June Huh — Hodge theory, combinatorics, matroids; James Maynard — analytic number theory, prime numbers; Maryna Viazovska — sphere packing, E_8 lattice, Fourier analysis	ICM 2022 was virtual; the award ceremony was held in Helsinki. Viazovska was the second woman Fields Medalist.

Essential ICM and Fields Medal Landmarks

- ICM stands for International Congress of Mathematicians.
- IMU stands for International Mathematical Union.
- The Fields Medal is awarded every four years at the ICM.
- The Fields Medal is named after John Charles Fields.
- The Fields Medal was first awarded in 1936.
- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal.
- Andrew Wiles received a special IMU silver plaque in 1998 for his proof of Fermat's Last Theorem.
- Grigori Perelman declined the Fields Medal in 2006 after proving the Poincaré conjecture.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- The 2026 ICM is scheduled to be held in Philadelphia, USA.

Part III

Past Problems

Problem 2014.1. This twentieth-century German mathematician is well known for a theorem stating that symmetries of Hamiltonian dynamical systems correspond to conservation laws. What is the name of this mathematician?

Solution. The theorem described in the problem is Noether's theorem. It states, roughly speaking, that continuous symmetries of a physical system correspond to conserved quantities. For example, time-translation symmetry corresponds to conservation of energy, and spatial-translation symmetry corresponds to conservation of momentum.

The mathematician associated with this theorem is Emmy Noether, one of the most important algebraists of the twentieth century.

Therefore, the answer is Emmy Noether.

□

Problem 2014.2. In the famous Chinese mathematical classic *The Nine Chapters on the Mathematical Art*, often compared with Euclid's *Elements* as a foundational mathematical classic, the value of π appears as 3. Later, Liu Hui, who wrote a commentary on *The Nine Chapters*, obtained the approximation 3.14 by refining the method of inscribed polygons. Which ancient Chinese mathematician further refined this method and obtained the approximations

$$\frac{355}{113} \quad \text{and} \quad \frac{22}{7}$$

for π ?

Solution. The mathematician described in the problem is Zu Chongzhi.

Liu Hui improved the approximation of π by using polygons with many sides. Zu Chongzhi later obtained the very accurate bounds

$$3.1415926 < \pi < 3.1415927,$$

and gave two famous rational approximations:

$$\frac{22}{7} \quad \text{and} \quad \frac{355}{113}.$$

The approximation $\frac{355}{113}$ is especially accurate and is traditionally known as the Milü, or "accurate ratio."

Therefore, the answer is Zu Chongzhi.

□

Problem 2014.3. Consider the 2014×2014 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}$ be the 2014×2014 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J} - \mathbf{I}.$$

The matrix $\mathbf{J}$ has eigenvalue 2014 for the vector

$$\mathbf{1} = (1, 1, \dots, 1)^T,$$

since

$$\mathbf{J}\mathbf{1} = 2014\mathbf{1}.$$

Therefore,

$$\mathbf{A}\mathbf{1} = (\mathbf{J} - \mathbf{I})\mathbf{1} = 2014\mathbf{1} - \mathbf{1} = 2013\mathbf{1}.$$

Hence 2013 is an eigenvalue of $\mathbf{A}$.

Now take any vector $\mathbf{v} = (v_1, \dots, v_{2014})^T$ such that

$$v_1 + \dots + v_{2014} = 0.$$

Then $\mathbf{J}\mathbf{v} = \mathbf{0}$, so

$$\mathbf{A}\mathbf{v} = (\mathbf{J} - \mathbf{I})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2013, so the multiplicity of -1 is 2013.

Therefore, the eigenvalues are

2013 with multiplicity 1,	-1 with multiplicity 2013.
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□

Problem 2014.4. Let $m \geq 3$ and $n \geq 3$. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq \mathbb{R}^m, \quad \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} \subseteq \mathbb{R}^n.$$

Assume that

$$V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

is a 2-dimensional subspace of $\mathbb{R}^m$, and that

$$W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$$

is a 3-dimensional subspace of $\mathbb{R}^n$. What is the rank of the $m \times n$ matrix

$$\mathbf{A} = \mathbf{v}_1\mathbf{w}_1^T + \mathbf{v}_2\mathbf{w}_2^T + \mathbf{v}_3\mathbf{w}_3^T?$$

Solution. Let

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 \end{pmatrix}.$$

Then

$$\mathbf{A} = \mathbf{P}\mathbf{Q}^T.$$

Since $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ span a 2-dimensional subspace, we have

$$\text{rank}(\mathbf{P}) = 2.$$

Since $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ span a 3-dimensional subspace, they are linearly independent, and hence

$$\text{rank}(\mathbf{Q}) = 3.$$

Therefore, the linear map $\mathbf{Q}^T : \mathbb{R}^n \rightarrow \mathbb{R}^3$ is surjective.

It follows that the image of $\mathbf{A} = \mathbf{P}\mathbf{Q}^T$ is

$$\text{Im}(\mathbf{A}) = \mathbf{P}(\text{Im}(\mathbf{Q}^T)) = \mathbf{P}(\mathbb{R}^3) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}.$$

This space has dimension 2. Hence

$$\text{rank}(\mathbf{A}) = 2.$$

Therefore, the answer is $\boxed{2}$.

□

Problem 2018.1. The following people are this year's Fields Medalists. Excluding choice (5), who is the second oldest among them?

- (1) Caucher Birkar
- (2) Alessio Figalli
- (3) Peter Scholze
- (4) Akshay Venkatesh
- (5) The person who stole Birkar's medal

Solution. The 2018 Fields Medalists were

- | | | |
|-----|------------------|---------------|
| (1) | Caucher Birkar | born in 1978, |
| (2) | Alessio Figalli | born in 1984, |
| (3) | Peter Scholze | born in 1987, |
| (4) | Akshay Venkatesh | born in 1981. |

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

(4).

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal. □

Problem 2018.2. According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of n possible geometric structures. What is n ?

Solution. The 3-dimensional geometric structures appearing in Thurston's geometrization conjecture are the eight Thurston geometries:

$$S^3, \quad \mathbb{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}_2(\mathbb{R})}, \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence $n = 8$, so the answer is

8.

□

Problem 2018.3. Find the sum of the following infinite series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots.$$

Solution. The given series is one half of the alternating harmonic series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right).$$

Recall the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

Substituting $x = 1$, we obtain

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

Therefore,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{\ln 2}{2}.$$

Thus the answer is $\boxed{\frac{\ln 2}{2}}$.

□

Problem 2018.4. Among the following five Fields Medalists, which one does not have a Korean doctoral student?

- (1) William Thurston
- (2) Maxim Kontsevich
- (3) Grigory Margulis
- (4) Terence Tao
- (5) Efim Zelmanov

Solution. The intended answer is Maxim Kontsevich.

This problem is not asking for a theorem but for advisor–student data about Fields Medalists and Korean mathematicians. In a Science Quiz setting, it is best treated as a memorization-style question: among the five listed Fields Medalists, the exceptional name in the intended answer key is

$\boxed{(2) \text{ Maxim Kontsevich}}$.

□

Problem 2020.1. The International Congress of Mathematicians, abbreviated as ICM, is a congress for mathematicians from around the world. It is organized every four years by the International Mathematical Union and may be thought of as a kind of Olympic Games for mathematicians. The 2014 ICM was held in South Korea, and the 2018 ICM was held in Brazil. As of 2019, in which country was the 2022 ICM scheduled to be held?

Solution. ICM stands for the International Congress of Mathematicians. It is the main international congress in mathematics and is organized every four years by the International Mathematical Union.

The 2014 ICM was held in Seoul, South Korea, and the 2018 ICM was held in Rio de Janeiro, Brazil. At the time this problem was asked in 2019, the 2022 ICM was scheduled to be held in Saint Petersburg, Russia. Therefore, the answer is $\boxed{\text{Russia}}$.

In reality, after Russia's invasion of Ukraine in 2022, the 2022 ICM was not held in Saint Petersburg. It was eventually held as a virtual event, with the award ceremony taking place in Helsinki, Finland. Thus, the answer to this problem should be understood according to the planned host country at the time of the 2019 competition.

□

Problem 2020.2. A fair six-sided die has the numbers $-2, 0, 2, 3, 4, 5$ written on its faces. The die is rolled 10 times, and X is defined to be the product of the 10 numbers obtained. What is the expected value of X ?

Solution. Let Y be the number obtained from one roll of the die. Since all six faces are equally likely,

$$\mathbb{E}[Y] = \frac{-2 + 0 + 2 + 3 + 4 + 5}{6} = 2.$$

Now let $Y_1, Y_2, \dots, Y_{10}$ be the numbers obtained from the 10 rolls. Then

$$X = Y_1 Y_2 \cdots Y_{10}.$$

Since the rolls are independent, the random variables $Y_1, Y_2, \dots, Y_{10}$ are independent. Therefore,

$$\mathbb{E}[X] = \mathbb{E}[Y_1 Y_2 \cdots Y_{10}] = \mathbb{E}[Y_1] \mathbb{E}[Y_2] \cdots \mathbb{E}[Y_{10}].$$

All the Y_i have the same distribution as Y , so

$$\mathbb{E}[X] = (\mathbb{E}[Y])^{10} = 2^{10} = 1024.$$

Thus the answer is 1024.

□

Problem 2020.3. Which of the following Fields Medalists was not serving as a professor at the time?

- (1) Cédric Villani
- (2) Elon Lindenstrauss
- (3) Peter Scholze
- (4) Martin Hairer
- (5) Terence Tao

Solution. At the time this problem was asked, Cédric Villani was active as a politician in France rather than serving as a professor.

The other choices were mathematicians holding professorial positions. Therefore, the answer is (1) Cédric Villani.

□

Problem 2020.4. In a three-dimensional gravitational field, when an object moves from a point $\mathbf{a}$ to another point $\mathbf{b}$, the change in its potential energy is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$. Which of the following mathematical facts is least directly related to this physical phenomenon?

- (1) A curl-free vector field is the gradient of some potential function.
- (2) Stokes' theorem
- (3) The Laplacian of $\frac{1}{|\mathbf{x}|}$ is a delta function, up to a constant factor.
- (4) The curl of a gradient is identically zero.

Solution. The statement that the change in potential energy is independent of the path means that the gravitational field is a conservative vector field.

Let $\mathbf{F}$ be the force field. The change in potential energy from $\mathbf{a}$ to $\mathbf{b}$ is given by

$$\Delta U = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{r}.$$

For this quantity to be independent of the path, the force field should be expressible as the negative gradient of a potential function:

$$\mathbf{F} = -\nabla U.$$

Thus choice (1) is directly related to the phenomenon.

Also, the curl of a gradient is always zero:

$$\nabla \times \nabla U = \mathbf{0}.$$

Hence choice (4) is also directly related. Stokes' theorem says that

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS.$$

Therefore, if $\nabla \times \mathbf{F} = \mathbf{0}$, then the line integral around a closed curve is zero, which is another way to understand path independence. Thus choice (2) is also related.

On the other hand,

$$\Delta \left(\frac{1}{|\mathbf{x}|} \right) = -4\pi\delta(\mathbf{x})$$

is related to the fundamental solution of the Laplacian and to Poisson's equation for gravitational potential. Although it is related to gravity, it is not the main mathematical reason why the potential energy change is independent of path. Therefore, the answer is (3).

□

Problem 2020.5. Among the following mathematics-related academic institutions, which one is actually a degree-granting school?

- (1) Institut des Hautes Études Scientifiques, IHES
- (2) École normale supérieure
- (3) Mathematical Sciences Research Institute, MSRI
- (4) Institute for Advanced Study, IAS
- (5) Collège de France

Solution. École normale supérieure is one of France's most prestigious higher education institutions. It operates educational programs and is naturally regarded as a degree-granting school.

By contrast, IHES, MSRI, and IAS are research institutes rather than degree-granting schools. Collège de France is also a famous academic institution that offers public lectures, but it does not grant degrees.

Therefore, the answer is (2) École normale supérieure.

□

Problem 2021.1. Which is larger, A or B , where

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{4}}}, \quad B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}?$$

Solution. First, we compute A :

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{4}}} = 1 + \frac{1}{1 + \frac{4}{5}} = 1 + \frac{1}{1 + \frac{5}{4}} = 1 + \frac{1}{\frac{9}{4}} = 1 + \frac{4}{9} = \frac{13}{9}.$$

Next, we compute B :

$$B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}} = 1 + \frac{1}{1 + \frac{3}{4}} = 1 + \frac{1}{1 + \frac{4}{5}} = 1 + \frac{1}{\frac{9}{5}} = 1 + \frac{5}{9} = \frac{14}{9}.$$

Since

$$\frac{13}{9} < \frac{14}{9},$$

we have $A < B$. Therefore, the answer is B.

□

Problem 2021.2. Let

$$A(n) = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^{n+1}.$$

What is the integer closest to $A(7)$?

Solution. Recall Binet's formula for the Fibonacci sequence:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Here

$$A(7) = \frac{\varphi^8}{\sqrt{5}}.$$

By Binet's formula,

$$F_8 = \frac{\varphi^8 - \psi^8}{\sqrt{5}}.$$

Since $|\psi| < 1$, the term $\frac{\psi^8}{\sqrt{5}}$ is very small. Thus

$$A(7) = \frac{\varphi^8}{\sqrt{5}} = F_8 + \frac{\psi^8}{\sqrt{5}}$$

is very close to F_8 .

The Fibonacci numbers are

$$F_0 = 0, \quad F_1 = 1, \quad F_2 = 1, \quad F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21.$$

Therefore, the integer closest to $A(7)$ is $\boxed{21}$.

□

Problem 2021.3. Choose all groups among the following that are not finitely generated:

- (1) The fundamental group of an open unit disk with finitely many punctures.
- (2) $\mathrm{SL}_2(\mathbb{Z})$, the group of 2×2 integer matrices with determinant 1.
- (3) $\mathbb{C}^\times$, the group of nonzero complex numbers under multiplication.
- (4) The Monster group, the largest sporadic finite simple group.

Solution. First, consider an open unit disk with finitely many punctures. If the removed points are $p_1, \dots, p_n$, then the fundamental group is isomorphic to the free group on n generators:

$$\pi_1(D \setminus \{p_1, \dots, p_n\}) \cong F_n.$$

Hence the group in choice (1) is finitely generated.

The group $\mathrm{SL}_2(\mathbb{Z})$ is also finitely generated. For example, it is generated by the two matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus choice (2) is finitely generated.

On the other hand, $\mathbb{C}^\times$ is an uncountable abelian group. Every finitely generated group is countable, so $\mathbb{C}^\times$ cannot be finitely generated. Therefore, choice (3) is not finitely generated.

Finally, the Monster group is a finite group. Every finite group is finitely generated, since one may simply take all of its elements as generators. Hence choice (4) is finitely generated.

Therefore, the only group in the list that is not finitely generated is $\boxed{(3) \mathbb{C}^\times}$.

□

Problem 2021.4. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find all eigenvalues of $\mathbf{A}$ together with their multiplicities. Also determine whether $\mathbf{A}$ is diagonalizable.

Solution. We can write $\mathbf{A}$ as

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$

Thus $\mathbf{A}$ is a rank-one matrix.

First, let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 6\mathbf{v}.$$

Hence 6 is an eigenvalue of $\mathbf{A}$.

Since $\mathbf{A}$ has rank 1, its nullity is 2. Therefore, the eigenspace corresponding to the eigenvalue 0 has dimension 2. Indeed,

$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ x + 2y + 3z \\ x + 2y + 3z \end{pmatrix},$$

so

$$E_0 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y + 3z = 0 \right\}.$$

Thus the geometric multiplicity of 0 is 2.

Now

$$\text{tr}(\mathbf{A}) = 1 + 2 + 3 = 6.$$

Since the sum of the eigenvalues, counted with algebraic multiplicity, is equal to the trace, the eigenvalues are

$$6, \quad 0, \quad 0.$$

Hence the algebraic multiplicities are

$$6 : 1, \quad 0 : 2.$$

Finally, the eigenspace for 6 has dimension 1, and the eigenspace for 0 has dimension 2. The total dimension of the eigenspaces is therefore

$$1 + 2 = 3.$$

Hence there exists a basis of $\mathbb{R}^3$ consisting of eigenvectors of $\mathbf{A}$. Therefore, $\mathbf{A}$ is diagonalizable.

Thus the eigenvalues are $\boxed{6, 0, 0}$, with algebraic multiplicities 1 and 2 respectively, and

$\boxed{\mathbf{A} \text{ is diagonalizable}}.$

□

Problem 2022.1. Which of the following numbers is the largest?

- (1) e^π
- (2) π^e
- (3) 7π
- (4) $\binom{7}{2}$
- (5) $4!$

Solution. We compare the values directly:

$$e^\pi \approx 23.1407, \quad \pi^e \approx 22.4592, \quad 7\pi \approx 21.9911.$$

Also,

$$\binom{7}{2} = \frac{7 \cdot 6}{2} = 21, \quad 4! = 24.$$

Therefore, the largest number among the choices is $\boxed{(5) \ 4!}$.

□

Problem 2022.2. Three people store a jointly owned treasure in a safe and want to lock it. They want no single person to be able to open the safe alone, but any two or more people should always be able to open it together. This can be done by putting n locks on the safe, making m copies of the key for each lock, and distributing the keys appropriately among the three people. Find n and m so that the number of locks n is minimized.

Solution. Let the three people be A , B , and C .

Since any two people should be able to open the safe, the key to each lock must be given to at least two people. If a certain lock had its key given to only one person, then the other two people would not be able to open that lock together.

Thus, for each lock, exactly one of the three people should not receive the key. On the other hand, no single person should be able to open the safe alone. Therefore, each person must be missing the key to at least one lock.

Hence we need at least three locks: one lock whose key is not given to A , one lock whose key is not given to B , and one lock whose key is not given to C . This lower bound is achievable by distributing the keys as follows:

	A	B	C
L_A	$\times$	$\checkmark$	$\checkmark$
L_B	$\checkmark$	$\times$	$\checkmark$
L_C	$\checkmark$	$\checkmark$	$\times$

That is, the key to L_A is given to B and C , the key to L_B is given to A and C , and the key to L_C is given to A and B .

Then each person is unable to open one of the locks, so no one can open the safe alone. However, any two people together can open all three locks. Therefore, the minimum is achieved by $\boxed{n = 3, \ m = 2}$.

□

Problem 2022.3. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2 + 7} dx.$$

Solution. Let

$$u = x - 2.$$

Then the limits of integration remain $-\infty$ and ∞ , so

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2+7} dx = 3 \int_{-\infty}^{\infty} \frac{1}{u^2+7} du.$$

We use the standard formula

$$\int_{-\infty}^{\infty} \frac{1}{u^2+a^2} du = \frac{\pi}{a}.$$

Here $a = \sqrt{7}$, and therefore

$$3 \int_{-\infty}^{\infty} \frac{1}{u^2+7} du = 3 \cdot \frac{\pi}{\sqrt{7}} = \frac{3\pi}{\sqrt{7}}.$$

Thus the answer is $\boxed{\frac{3\pi}{\sqrt{7}}}$.

□

Problem 2022.4. A couple has two children.

- (a) Given that one of the children is a daughter, what is the probability that the other child is also a daughter?
- (b) Given that one of the children is a daughter named Youngmi, what is the probability that the other child is also a daughter?

Solution. Let G denote a girl and B denote a boy. If we distinguish the two children by birth order, then the possible gender combinations are

$$GG, \quad GB, \quad BG, \quad BB.$$

For part (a), the condition that one child is a daughter means that at least one child is a girl. Thus the possible cases are

$$GG, \quad GB, \quad BG.$$

Among these three cases, only GG has the property that the other child is also a daughter. Therefore,

$$\mathbb{P}(\text{both are daughters} \mid \text{at least one is a daughter}) = \frac{1}{3}.$$

For part (b), the information that one child is a daughter named Youngmi is more specific. In the intended interpretation of the problem, this information identifies one particular child. Then the gender of the other child is still equally likely to be G or B . Hence

$$\mathbb{P}(\text{the other child is a daughter}) = \frac{1}{2}.$$

Therefore, the answers are $\boxed{\text{(a) } \frac{1}{3}, \text{ (b) } \frac{1}{2}}$.

□

Problem 2022.5. Which of the following conjectures was not solved by Professor June Huh?

- (1) Read's conjecture
- (2) Brylawski's conjecture
- (3) Poincaré conjecture
- (4) Rota's conjecture
- (5) Okounkov's conjecture

Solution. Professor June Huh is known for connecting combinatorics with algebraic geometry and using these ideas to solve several major conjectures in combinatorics. His work is related to conjectures such as Read's conjecture, Brylawski's conjecture, Rota's conjecture, and Okounkov's conjecture.

On the other hand, the Poincaré conjecture is a famous problem in 3-dimensional topology. It was proved by Grigori Perelman using Ricci flow, not by June Huh.

Therefore, the conjecture in the list that was not solved by June Huh is

(3) Poincaré conjecture.

□

Problem 2022.6. This mathematician, often called the father of linear programming, is famous for solving difficult problems while he was a doctoral student. One day, while studying at UC Berkeley, he arrived late to a class. The statistics professor Jerzy Neyman had written two famous unsolved problems on the blackboard, but the student thought they were homework problems. He took them home, solved them, and submitted complete solutions a few days later, thinking only that he had turned in the homework late. Who was this mathematician?

Solution. The mathematician described in the problem is George Dantzig.

George Dantzig is known as one of the founders of linear programming and as the creator of the simplex method. For this reason, he is often called the father of linear programming.

The anecdote in the problem is also a famous story about Dantzig. While he was a doctoral student at UC Berkeley, he arrived late to Jerzy Neyman's statistics class and copied down two problems written on the blackboard, believing that they were homework problems. He later submitted solutions, not realizing that the problems were actually open problems in statistics.

Therefore, the answer is George Dantzig.

□

Problem 2022.7. Two different coins are each tossed three times. Let X and Y be the numbers of heads obtained from the two coins, respectively. What is the probability that $X = Y$?

Solution. For each coin, the number of heads obtained in three tosses follows the binomial distribution

$$X, Y \sim \text{Bin}\left(3, \frac{1}{2}\right).$$

Also, X and Y are independent. Therefore,

$$\mathbb{P}(X = Y) = \sum_{k=0}^3 \mathbb{P}(X = k) \mathbb{P}(Y = k).$$

For each $k = 0, 1, 2, 3$, we have

$$\mathbb{P}(X = k) = \binom{3}{k} \left(\frac{1}{2}\right)^3,$$

and the same formula holds for Y . Hence

$$\mathbb{P}(X = Y) = \sum_{k=0}^3 \left(\binom{3}{k} \left(\frac{1}{2}\right)^3 \right)^2 = \frac{1}{64} \sum_{k=0}^3 \binom{3}{k}^2.$$

Thus

$$\mathbb{P}(X = Y) = \frac{1}{64} \left(\binom{3}{0}^2 + \binom{3}{1}^2 + \binom{3}{2}^2 + \binom{3}{3}^2 \right) = \frac{1 + 9 + 9 + 1}{64} = \frac{20}{64} = \frac{5}{16}.$$

Therefore, the answer is $\boxed{\frac{5}{16}}$.

□

Problem 2023.1. In which year was the Fields Medal, won by Professor June Huh, first awarded?

Solution. The Fields Medal is one of the most prestigious awards in mathematics, and it is awarded by the International Mathematical Union. Professor June Huh received the Fields Medal in 2022.

The Fields Medal was first awarded in 1936. The first two recipients were Lars Ahlfors and Jesse Douglas.

Therefore, the answer is $\boxed{1936}$.

□

Problem 2023.2. Consider the 6×6 square grid with vertices $(0, 0)$, $(6, 0)$, $(0, 6)$, and $(6, 6)$. A path starts at $(0, 0)$ and ends at $(6, 6)$, moving only one unit to the right or one unit upward at each step. How many such paths never go above the line $x = y$? Equivalently, how many such paths stay in the region

$$\{(x, y) \in \mathbb{R}^2 : y \leq x\}?$$

Solution. To go from $(0, 0)$ to $(6, 6)$, a path must consist of 6 right steps and 6 upward steps. Without the condition $y \leq x$, the total number of such paths is

$$\binom{12}{6}.$$

We now count the paths that do go above the line $y = x$. By the reflection principle, the number of paths from $(0, 0)$ to $(6, 6)$ that at some point satisfy $y > x$ is

$$\binom{12}{5}.$$

Therefore, the number of paths that always stay in the region $y \leq x$ is

$$\binom{12}{6} - \binom{12}{5}.$$

Computing this gives

$$\binom{12}{6} - \binom{12}{5} = 924 - 792 = 132.$$

Thus the answer is $\boxed{132}$.

□

Problem 2023.3. A point on the surface of the Earth has the following property: starting from that point, one moves 5km south, then 5km east, and then 5km north, and returns to the original point. How many such points are there on the surface of the Earth?

Assume that the surface of the Earth is a perfect two-dimensional sphere, and that the North Pole and the South Pole are antipodal points. At the North Pole, the directions north, east, and west are not defined, and at the South Pole, the directions south, east, and west are not defined. Starting points for which one of these undefined directions occurs during the movement are excluded.

Solution. The North Pole is one such point. If we start at the North Pole, move 5km south, then move 5km east along a latitude circle, and finally move 5km north, we return to the North Pole.

There are also many such points near the South Pole. Let Q be the point reached after moving 5km south from the starting point. Suppose that the latitude circle passing through Q has circumference

$$\frac{5}{n} \text{ km}$$

for some positive integer n . Then moving 5km east from Q goes exactly n full times around that latitude circle and returns to Q . After that, moving 5km north returns to the original starting point.

Near the South Pole, the circumferences of latitude circles approach 0. Hence, for every positive integer n , there is a latitude circle near the South Pole whose circumference is exactly $\frac{5}{n}$ km. For each such latitude circle, every point lying 5km north of it gives a valid starting point.

Therefore, there are not just one but infinitely many such points on the Earth's surface. In fact, this construction gives entire latitude circles of valid starting points, so there are uncountably many such points. For the usual quiz answer, the essential conclusion is

infinitely many.

□

Problem 2023.4. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a continuous function. For each $\lambda > 0$, define

$$f_\lambda(x) := \lambda f\left(\frac{x}{\lambda}\right),$$

where f_λ is defined for $x \in (-\lambda, \lambda)$.

Suppose that for every sequence $\lambda_i \rightarrow \infty$, the functions f_{λ_i} converge locally uniformly to 0 on $\mathbb{R}$. Determine whether the following statement is true or false:

$$f \text{ is differentiable at } 0 \text{ and } f'(0) = 0.$$

Solution. We prove that the statement is true.

First, evaluate at $x = 0$. For every sequence $\lambda_i \rightarrow \infty$, we have

$$f_{\lambda_i}(0) = \lambda_i f(0).$$

Since f_{λ_i} converges locally uniformly to 0, in particular it converges pointwise to 0 at $x = 0$. Hence

$$\lambda_i f(0) \rightarrow 0.$$

This is possible for all sequences $\lambda_i \rightarrow \infty$ only if

$$f(0) = 0.$$

Now we show that $f'(0) = 0$. Let $h_i \rightarrow 0$ be any sequence with $h_i \neq 0$. For all sufficiently large i , we have $h_i \in (-1, 1)$. Define

$$\lambda_i = \frac{1}{|h_i|}.$$

Then $\lambda_i \rightarrow \infty$, and

$$h_i = \frac{\text{sgn}(h_i)}{\lambda_i}.$$

By the assumption, $f_{\lambda_i} \rightarrow 0$ locally uniformly on $\mathbb{R}$. In particular, this convergence is uniform on the compact set $\{-1, 1\}$. Therefore,

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) \rightarrow 0.$$

But

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) = \lambda_i f\left(\frac{\operatorname{sgn}(h_i)}{\lambda_i}\right) = \lambda_i f(h_i).$$

Hence

$$\lambda_i f(h_i) \rightarrow 0.$$

Since $f(0) = 0$, we get

$$\left| \frac{f(h_i) - f(0)}{h_i} \right| = \left| \frac{f(h_i)}{h_i} \right| = \lambda_i |f(h_i)| \rightarrow 0.$$

Thus

$$\frac{f(h_i) - f(0)}{h_i} \rightarrow 0$$

for every sequence $h_i \rightarrow 0$ with $h_i \neq 0$. Therefore, f is differentiable at 0 and

$$f'(0) = 0.$$

Thus the answer is True.

□

Problem 2024.1. When all natural numbers from 100 to 999 are written in a row, how many times does the digit 0 appear?

Solution. Every natural number from 100 to 999 is a three-digit number.

In a three-digit number, the hundreds digit cannot be 0. Therefore, the digit 0 can appear only in the tens digit or the units digit.

First, count the numbers whose tens digit is 0. The hundreds digit can be one of $1, 2, \dots, 9$, and the units digit can be one of $0, 1, \dots, 9$. Hence there are

$$9 \cdot 10 = 90$$

such numbers.

Next, count the numbers whose units digit is 0. The hundreds digit can be one of $1, 2, \dots, 9$, and the tens digit can be one of $0, 1, \dots, 9$. Hence there are again

$$9 \cdot 10 = 90$$

such numbers.

Therefore, the total number of appearances of the digit 0 is

$$90 + 90 = 180.$$

Thus the answer is 180.

□

Problem 2024.2. Which of the following people was born the latest?

- (1) John von Neumann
- (2) Alan Turing
- (3) John Nash
- (4) Pierre de Fermat

Solution. We compare their years of birth:

- (1) John von Neumann: born in 1903,
- (2) Alan Turing: born in 1912,
- (3) John Nash: born in 1928,
- (4) Pierre de Fermat: born in 1607.

Among these four, the person born the latest is John Nash.

Therefore, the answer is (3) John Nash.

□

Problem 2024.4. Let C be a convex subset of $\mathbb{R}^3$. Here convex means that for any two points $\mathbf{p}, \mathbf{q} \in C$, the line segment joining $\mathbf{p}$ and $\mathbf{q}$ is also contained in C .

Suppose that the following two facts are known:

- (i) The intersection of C with the xy -plane is

$$C \cap \{z = 0\} = \{(x, y, 0) : x^2 + 4y^2 \leq 1\}.$$

- (ii) The set C contains the z -axis.

Can the volume of C in the region $-1 \leq z \leq 1$ be determined?

- (1) This cannot be determined.
- (2) π
- (3) 2π
- (4) 4π

Solution. Let

$$E = \{(x, y) \in \mathbb{R}^2 : x^2 + 4y^2 \leq 1\}.$$

By condition (i),

$$C \cap \{z = 0\} = E \times \{0\}.$$

The ellipse E has semiaxes 1 and $\frac{1}{2}$, so its area is

$$\text{area}(E) = \pi \cdot 1 \cdot \frac{1}{2} = \frac{\pi}{2}.$$

For each height z , define the horizontal cross-section

$$C_z = \{(x, y) \in \mathbb{R}^2 : (x, y, z) \in C\}.$$

We claim that C_z has the same area as E for every z .

First, we show that $C_z \subseteq E$. Suppose that $(x, y, z) \in C$ and

$$x^2 + 4y^2 > 1.$$

Since C contains the z -axis, the point $(0, 0, t)$ belongs to C for every $t \in \mathbb{R}$. By convexity, the line segment joining (x, y, z) and $(0, 0, t)$ is contained in C .

For any $\alpha \in (0, 1)$, we may choose

$$t = -\frac{\alpha z}{1 - \alpha}$$

so that this segment intersects the xy -plane at the point

$$(\alpha x, \alpha y, 0).$$

Since α can be chosen sufficiently close to 1, we may arrange that

$$(\alpha x)^2 + 4(\alpha y)^2 > 1.$$

This would give a point of $C \cap \{z = 0\}$ lying outside $E \times \{0\}$, contradicting condition (i). Therefore,

$$C_z \subseteq E$$

for every z .

Conversely, we show that the interior of E is contained in every C_z . Take (x, y) with

$$x^2 + 4y^2 < 1.$$

Choose $\alpha \in (0, 1)$ sufficiently close to 1 so that

$$\left(\frac{x}{\alpha}\right)^2 + 4\left(\frac{y}{\alpha}\right)^2 \leq 1.$$

Then

$$\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) \in C.$$

Also, since C contains the z -axis,

$$\left(0, 0, \frac{z}{1-\alpha}\right) \in C.$$

By convexity, their convex combination also belongs to C , and

$$\alpha \left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) + (1-\alpha) \left(0, 0, \frac{z}{1-\alpha}\right) = (x, y, z).$$

Hence $(x, y) \in C_z$.

Thus, for every z ,

$$\text{int}(E) \subseteq C_z \subseteq E.$$

Since the boundary of the ellipse has area 0, we have

$$\text{area}(C_z) = \text{area}(E) = \frac{\pi}{2}$$

for every z . Therefore, the volume of C in the region $-1 \leq z \leq 1$ is

$$\int_{-1}^1 \text{area}(C_z) dz = \int_{-1}^1 \frac{\pi}{2} dz = \pi.$$

Therefore, the answer is $\boxed{(2) \pi}$.

□

Problem 2025.1. A fair coin is tossed repeatedly until either two heads in a row or two tails in a row appear. On average, how many tosses are needed before the process stops?

Solution. On the first toss, it is impossible to have two consecutive equal outcomes. After the first toss, the process stops exactly when the next toss gives the same result as the previous toss.

Since the coin is fair, from the second toss onward we have

$$\mathbb{P}(\text{the new toss matches the previous toss}) = \frac{1}{2}.$$

Therefore, after the first toss, the number of additional tosses needed until the process stops follows a geometric distribution with success probability $\frac{1}{2}$. Its expected value is

$$\frac{1}{1/2} = 2.$$

Since the first toss is always made, the expected total number of tosses is

$$1 + 2 = 3.$$

Thus the answer is 3.

□

Problem 2025.2. The nineteenth-century mathematicians Cauchy and Weierstrass established rigorous definitions of limits. Who was the first mathematician to prove that the sum of the reciprocals of the squares of all positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}?$$

- (1) David Hilbert
- (2) Georg Cantor
- (3) Bernhard Riemann
- (4) Leonhard Euler

Solution. The problem of finding the exact value of the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is known as the Basel problem.

This problem had been known since the seventeenth century, and it was Leonhard Euler who first found and proved the famous identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the answer is (4) Leonhard Euler.

□

Problem 2025.3. Find the equation of the smallest sphere tangent to all four planes

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y + z = 3.$$

Solution. Since the sphere is tangent to the three coordinate planes

$$x = 0, \quad y = 0, \quad z = 0,$$

and we are looking for the smallest such sphere, its center must lie in the first octant and have the form

$$\mathbf{c} = (a, a, a)$$

by symmetry. The distance from $\mathbf{c}$ to each coordinate plane is a , so the radius of the sphere is

$$r = a.$$

The sphere must also be tangent to the plane

$$x + y + z = 3.$$

The distance from (a, a, a) to this plane is

$$\frac{|a + a + a - 3|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{3 - 3a}{\sqrt{3}},$$

since the smallest sphere lies between the coordinate planes and the plane $x + y + z = 3$. This distance must equal the radius a , so

$$\frac{3 - 3a}{\sqrt{3}} = a.$$

Hence

$$3 - 3a = \sqrt{3}a,$$

and therefore

$$a = \frac{3}{3 + \sqrt{3}} = \frac{3 - \sqrt{3}}{2}.$$

Thus the center is

$$\left(\frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2} \right),$$

and the radius is

$$\frac{3 - \sqrt{3}}{2}.$$

Therefore, the equation of the sphere is

$$\left[\left(x - \frac{3 - \sqrt{3}}{2} \right)^2 + \left(y - \frac{3 - \sqrt{3}}{2} \right)^2 + \left(z - \frac{3 - \sqrt{3}}{2} \right)^2 = \left(\frac{3 - \sqrt{3}}{2} \right)^2 \right].$$

□

Part IV

Expected Problems